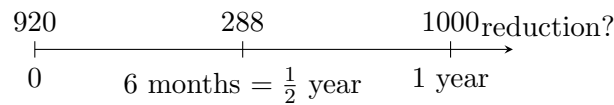


Problem 8.19

A signs a one-year note for \$1000 and receives \$920 from the bank. At the end of six months, A makes a payment of \$288. Assuming simple discount, to what amount does this reduce the face amount of the note?

Solution

The cash flows can be represented on a timeline as follows:



First, find the simple discount rate. Since A receives \$920 on a one-year note with face amount \$1000,

$$920 = 1000(1 - dt), \quad t = 1.$$

Thus,

$$\begin{aligned} 0.92 &= 1 - d, \\ d &= 0.08. \end{aligned}$$

Let F be the amount by which the face amount of the note is reduced by the payment of \$288 made after six months. Under simple discount, the payment is the discounted value of F for the remaining half year:

$$288 = F \left(1 - d \cdot \frac{1}{2} \right).$$

Substituting $d = 0.08$ gives

$$\begin{aligned} 288 &= F \left(1 - 0.08 \cdot \frac{1}{2} \right), \\ 288 &= F(0.96), \\ F &= \frac{288}{0.96}, \\ F &= 300. \end{aligned}$$

The payment reduces the face amount by \$300. Therefore, the new face amount of the note is

$$1000 - 300 = 700.$$

Thus, the face amount is reduced to

$$\boxed{\$700}.$$